RESEARCH CURRICULUM VITAE | FALL 2027

A.K.M. AMANULLAH

Mechanical Engineer • Lecturer, BUBT • Hydrogen, sustainable energy and thermal systems

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RESEARCH FOCUS

Prospective PhD researcher in energy conversion, waste-to-hydrogen, computational thermodynamics and thermal management. My work connects mechanistic process models with data-driven forecasting, optimization and uncertainty quantification, with an emphasis on transparent validation and practical sustainability decisions.

EDUCATION

M.Sc., Mechanical Engineering — Rajshahi University of Engineering & Technology (RUET) | *In progress, 2025–present*

Graduate coursework completed; thesis remaining. Part-time enrollment.

M.Sc., Mechanical Engineering — Bangladesh University of Engineering and Technology (BUET) | *Admission offered, 2026*

Admitted for the April 2026 semester; enrollment and progress not asserted.

B.Sc., Mechanical Engineering — RUET | *Graduated 2024*

CGPA: 3.51/4.00. Seventh-semester GPA: 3.86/4.00; ranked 3rd of 181 students in that semester.

PEER-REVIEWED JOURNAL PUBLICATION

A.K.M. Amanullah and Md. Rabiul Islam Sarker (2026). "Integrated hydrogen potential from municipal organic waste in Bangladesh: A comprehensive, uncertainty-quantified multi-model framework integrating ARIMA–XGBoost forecasting, Aspen Plus simulation, and comparative techno-economic analysis." *International Journal of Hydrogen Energy, 254, 156339.*

DOI: 10.1016/j.ijhydene.2026.156339

- First-author analysis combining 22-year municipal-waste records, ARIMA–XGBoost forecasting, Aspen Plus V12 process models (AD–SMR), Monte Carlo uncertainty assessment (10,000 iterations), Sobol sensitivity indices, techno-economic analysis and streamlined LCA.
- Published model outcomes include a projected 26.2 Mt H_2 (95% CI 21.4–31.7 Mt), LCOH of $3.80/kg and 2.1 kg CO_2-eq/kg H_2 cradle-to-gate carbon footprint; these are modeled, not pilot-plant results.

CONFERENCE PAPERS ACCEPTED — ICMIME 2026 (RUET)

9–11 December 2026 • accepted for conference publication as reported by the author; proceedings not yet released.

Paper 318 *Long-Term Anaerobic Co-Digestion of Cafeteria Vegetable Waste: Comparative Methane Concentration and pH Dynamics.* Minhazul Alom, A.K.M. Amanullah (corresponding author), et al.

Paper 331 *Comparative Assessment of Conventional Bleach and Ozone-Assisted Denim Washing: Resource Use, Fabric Performance, and Environmental Impact.* Naeem Ahmmed, A.K.M. Amanullah, et al.

Paper 338 *Comparative Assessment of Conventional and Alternative Stripping Processes for Reactive-Dyed Cotton Fabrics.* Eshita Datta, A.K.M. Amanullah (co-corresponding author), et al.

Paper 339 *Development and Preliminary Evaluation of a Bio-Based Multifunctional Finish for Cotton: Antibacterial Activity, Flame-Spread Behavior and Fabric Performance.* Naeem Ahmmed, A.K.M. Amanullah, et al.

ADDITIONAL RESEARCH MANUSCRIPTS

Thermoelectric-Assisted Direct Evaporative Cooling for Humid Tropical Conditions

Co-author, journal manuscript in preparation with Md. Emdadul Hoque and colleagues. Analyzed 12 prototype comparisons and 29 Box–Behnken simulation runs using RSM, model reconstruction and sensitivity analysis; mean reported assisted-versus-unassisted air-temperature reduction advantage was 2.63 °C.

Flow-Electrode Capacitive Deionization for Sustainable Water Treatment

Co-author, conference manuscript for ICMIE 2026 (acceptance/publication not independently confirmed). Semi-empirical performance modeling and Pareto analysis of salt-removal efficiency versus electrical demand; predicted >90% removal under selected model conditions.

ACADEMIC & PROFESSIONAL APPOINTMENTS

Lecturer | Department of Textile Engineering, BUBT | *2026–present*

Teach Fundamentals of Mechanical Engineering; Fundamentals of Machine Dynamics and Design; Automation and Robotics in Textiles; and Engineering Drawing Laboratory (courses listed in prior SOP).

Mechanical Engineering Instructor | BDIT & Language Institute | *June 2024–March 2026*

Delivered engineering theory and practical instruction, supported assessments, laboratory teaching and student advising.

Industrial Trainee | Ashuganj Power Station Company Limited | *2023*

Completed industrial training in a power-generation facility, 25 March–17 April 2023.

RESEARCH METHODS & SKILLS

Energy and simulation: Aspen Plus V12, thermodynamic/energy-system modeling, hydrogen conversion pathways, exergy and TEA/LCA interpretation.

Data science: Python, ARIMA, XGBoost, regression, rolling-origin evaluation, forecasting.

Optimization & robustness: Monte Carlo simulation, Sobol global sensitivity, Box–Behnken design, response-surface methodology, ANOVA, multi-objective/Pareto optimization.

Engineering tools: SolidWorks and ANSYS (academic exposure); thermodynamics, heat transfer, fluid mechanics, engineering drawing.

ACADEMIC SERVICE, AWARDS AND TRAINING

- BUBT Sustainable Development Goals (SDG) Team 2, primary SDG 4; secondary SDGs 11 and 12 (2026). University ranking/sustainability-related service as reported in prior CV.
- RUET technical scholarships (2020, 2021, 2022), and earlier board scholarships (as documented in previous CV).
- ICMIME 2022 volunteer recognition; Hult Prize RUET campus participation (2020).

STANDARDIZED ENGLISH PROFICIENCY

IELTS Academic 7.0 overall (Listening 7.0; Reading 7.0; Writing 6.5; Speaking 6.5), tested 11 October 2026.

RESEARCH THEMES

Waste-to-hydrogen and bioenergy • process systems engineering • uncertainty-aware optimization • renewable thermal systems and heat transfer • low-carbon conversion • water–energy nexus.